

Estimating Mutational Signatures on Evolutionary Trees using Hierarchical Dirichlet Processes

Lab rotation report on fixed-signature and de novo signature inference

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Abstract

Recovering mutational signature activities across tumour subclones is critical for understanding cancer evolution, but standard tools like Non-negative Matrix Factorisation (NMF) discard known phylogenetic relationships. To leverage evolutionary relatedness, we implemented a tree-structured Hierarchical Dirichlet Process (Tree-HDP) that models signature activities as a random walk down the phylogeny. However, sparse signature activities push the posterior toward the simplex boundary, creating narrow geometric funnels that stall standard Hamiltonian Monte Carlo (NUTS) samplers. We resolved this pathology by reparameterising the walk into an unconstrained logit space using a symmetric sum-zero link, enabling efficient, stable sampling. Evaluated on simulated cohorts, this logistic-normal walk eliminates sampler stalling and cluster multimodality. The tree-aware model recovers signature activities more accurately than NMF while the signatures remain distinguishable, and a larger forest extends that advantage to higher overlap, though at the strongest overlap the signatures become non-identifiable and neither method recovers them.

1 Introduction

A cancer genome accumulates somatic mutations from several distinct mutational processes. Each process leaves a characteristic fingerprint, a mutational signature, which is a probability distribution over the 96 trinucleotide substitution channels ($\theta \in \Delta^{96}$) [3]. The observed mutation spectrum of a tumour is a mixture of these signatures combined with different activities. Recovering which processes were active and how strongly is a central task in cancer genomics, since it points to the mechanisms driving the disease.

The standard tool for this is Non-negative Matrix Factorisation (NMF), which factors a count matrix $X \approx SE$ into signatures S and activities E [2, 8]. NMF treats samples as exchangeable, without the notion that some samples are more closely related than others. But single cell sequencing now resolves a tumour into subclones and lets upstream tools reconstruct the phylogenetic tree that relates them [6], and that tree is exactly the information NMF discards. Subclones sharing recent ancestry should have similar activity, so placing the activities on the tree lets a model share statistical strength between related subclones and recover them more reliably. The project therefore targets a tree aware model, Tree-HDP, that uses evolutionary relatedness as a prior structure.

2 The generative model

The simulator is a finite dimensional approximation of a tree structured Hierarchical Dirichlet Process [11]. Instead of a nested Dirichlet Process at each node, activities follow a Dirichlet random walk down the tree. For a fixed signature matrix S (K signatures by 96 channels) and a forest of trees, the simulator draws

$$e_0 \sim \text{Dir}\left(\frac{\alpha_0}{K} \mathbf{1}_K\right), \quad e_j \sim \text{Dir}(\alpha \cdot e_{\text{parent}(j)}), \quad M_j \sim \text{NegBin}(\lambda_j, r), \quad x_j \sim \text{Multinomial}(M_j, e_j S). \quad (1)$$

Here e_0 is a baseline cohort, $e_j \in \Delta^K$ the activity at node j (the root parent is e_0), M_j the mutation burden, and $x_j \in \mathbb{N}^{96}$ the observed channel counts. The burden M_j follows a negative binomial with mean λ_j and variance $\lambda_j(1 + \lambda_j/r)$, to account for over dispersion. The dispersion r is taken as $r = 2$ in the fixed-signature experiments and to the Poisson limit in the de novo ones.

The Dirichlet walk $e_j \sim \text{Dir}(\alpha e_{\text{parent}(j)})$ is a mean-preserving step on the simplex, since $\mathbb{E}[e_j | e_{\text{parent}(j)}] = e_{\text{parent}(j)}$, so each node is a noisy copy of its parent with no systematic drift (Figure 1). The concentration α sets the step size, with large α giving tight inheritance and small α giving loose, which formally encodes the expectation that related subclones have similar activity. The baseline concentration per component is α_0/K . Values below 1 give a sparse e_0 and values ≥ 1 a spread out one, so α_0 acts as a sparsity control.

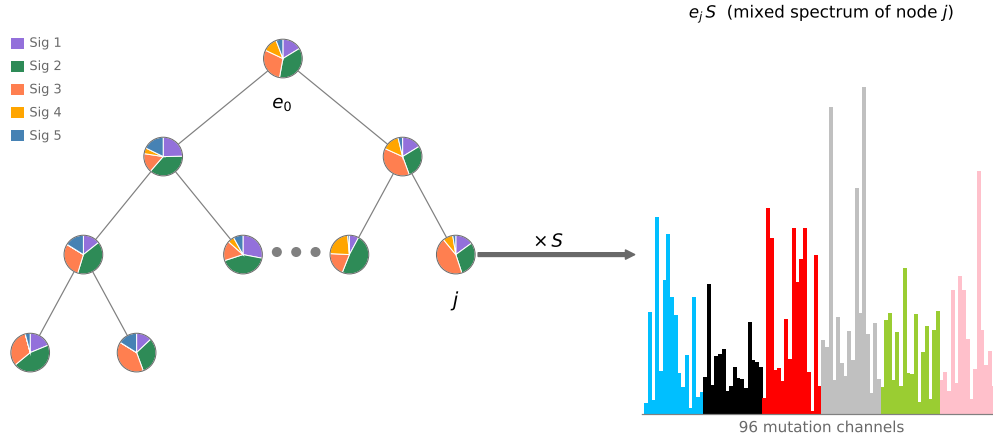


Figure 1: The activity walk on a forest with $K = 5$ signatures. All trees descend from a shared baseline e_0 , and each node, shown as a pie over the signatures, is a noisy copy of its parent, so activity drifts down each tree. For the leaf j it is mapped through S to the 96-channel spectrum $e_j S$ that the counts x_j are drawn from.

3 The fixed-signature model

The fixed-signature model treats the signature matrix S as known and infers only the per node activities $e_j \in \Delta^K$, which isolates activity inference from the harder discovery problem. It mirrors the generative process. Each node's activity follows the tree walk centred on its parent and the observed counts are multinomial. The natural choice, matching the generator, places the walk directly on the simplex through a Dirichlet step,

$$e_0 \sim \text{Dir}(\mathbf{1}_K), \quad e_j \sim \text{Dir}(\alpha e_{\text{parent}(j)}), \quad x_j \sim \text{Multinomial}(M_j, e_j S), \quad (2)$$

with a single shared concentration α under a $\text{LogNormal}(3, 0.5)$ prior and the root centred on a cohort baseline e_0 . It is the obvious model, but it is hard to sample.

Most nodes are active in only a few of the K signatures, so their activity vectors lie near the boundary of the simplex, where several components are almost zero. This is where the Dirichlet walk becomes hard to sample. A component that is close to zero has almost no room to move before it would leave the simplex, so the posterior is very thin in that direction while staying wide in the others. Which directions are thin changes from node to node, so the posterior is a narrow, curved valley whose width varies along its length creating a funnel.

NUTS [5] moves through the posterior in steps of a single size, tuned once for the whole space. A funnel has no good single step size. A step large enough to cross the wide part overshoots in the thin part and lands where the probability is negligible, which the sampler rejects as a divergence, while a step small enough for the thin part barely moves anywhere else. The chains therefore stall and the convergence diagnostics stay poor.

The fix is to change coordinates so that this geometry disappears. We parameterise the walk in unconstrained logit space and map to the simplex with a softmax, taking Gaussian increments down the tree,

$$\eta_{\text{root}} \sim \mathcal{N}(\mathbf{0}, \sigma_0^2 I), \quad \eta_j = \eta_{\text{parent}(j)} + \sigma z_j, \quad z_j \sim \mathcal{N}(\mathbf{0}, I), \quad e_j = \text{softmax}([\eta_j, 0]). \quad (3)$$

Here η_j and z_j are $(K - 1)$ -vectors, since the softmax with a pinned last logit leaves $K - 1$ free coordinates, so I is the $(K - 1) \times (K - 1)$ identity. The step scale σ has a $\text{LogNormal}(0, 1)$ prior and the root scale is fixed at $\sigma_0 = 2$. The last logit is pinned to 0 as the softmax identifiability anchor. In logit space a vanishing activity becomes a logit running to minus infinity rather than a hard edge at finite distance, so the sampler never pushes against a boundary and the space stays smooth near small activities. The walk is also written non-centred. Written directly, a node's logit is drawn around its parent with spread σ , so the width of the region the sampler explores changes with σ , a funnel that no single step size fits. Instead we sample a standard increment z_j of unit spread and form the logit as $\eta_{\text{parent}(j)} + \sigma z_j$, so the step scale is decoupled from the values and the funnel is gone. These Gaussian increments in logit space, mapped to the simplex by the softmax, are a logistic-normal walk, and the result is a smooth, well-conditioned posterior that NUTS, run in PyMC [1], samples with a single step size.

3.1 The fixed-signature results

We test the fixed-signature model on simulated forests with controlled ground truth. The signatures are ten fixed COSMIC SBS profiles [4] spanning a wide range of peakedness, the trees are shallow at three to seven nodes, activity is sparse so a typical node mixes only a few signatures, and per node mutation counts are negative binomial with mean about 970. Varying the number of trees over $\{5, 10, 15, 20, 25, 30, 35, 40\}$ with the signature set held fixed, we ask how convergence and accuracy change with the amount of data. Convergence is the worst $\hat{r}$ and the smallest bulk ESS across the activity parameters and σ [12]. Accuracy is the per node activity recovery against the truth, as the median L_1 distance to the true activities on the simplex, which runs from 0 to 2 and measures the total activity mass a node places on the wrong signatures.

The model recovers the activities accurately even with very little data. At the smallest forests, five trees, the median per node L_1 to the truth is already 0.17, and it does not fall systematically as the forest grows. Convergence is untroubled throughout, the worst $\hat{r}$ at 1.00 and the smallest bulk ESS never below about 1600 for 4 chains 2000 draws each, so adding trees does not strain the sampler (Table 1). All runs use NUTS in PyMC, and the model code, the configuration

files that reproduce every run are in the repository at https://github.com/RadoslavJochman/Mutational_signatures_HDP.

trees	max $\hat{r}$	min bulk ESS	median L_1
5	1.00	2259	0.17
10	1.00	2520	0.15
15	1.00	3353	0.15
20	1.00	3100	0.15
25	1.00	1825	0.09
30	1.00	1603	0.12
35	1.00	2231	0.17
40	1.00	2489	0.11

Table 1: Fixed-signature convergence and accuracy against the number of trees. Convergence is the worst split- $\hat{r}$ and the smallest bulk ESS over the activity variables and σ . Accuracy is the median per-node L_1 to the true activities, which runs from 0 to 2. Convergence stays clean and accuracy does not degrade as the forest grows.

With activity inference on a known signature set well behaved, the next step is to drop that assumption and infer the signatures jointly with the activities.

4 De novo signature discovery

In the fixed-signature model the signatures were supplied and only the per node activities were inferred. The de novo model makes the signature matrix latent and infers it jointly with the activities since the processes active in a given cohort need not match the COSMIC reference and may include profiles outside it.

4.1 The model

Each of the K signatures is given an uninformative Dirichlet prior over the 96 channels, and the activity walk and the count likelihood are as in the fixed-signature model.

$$S_k \sim \text{Dir}(\beta \cdot \mathbf{1}_{96}), \quad k = 1, \dots, K \quad (4)$$

$$\eta_j = \eta_{\text{parent}(j)} + \sigma z_j, \quad e_j = \text{softmax}([\eta_j, 0]) \quad (5)$$

$$x_j \sim \text{Multinomial}(M_j, e_j S) \quad (6)$$

A single shared concentration $\beta = 0.5$ governs every signature, which is a deliberate first simplification, and K is held at its true value.

4.2 Why de novo is harder

Making S latent introduces two non-identifiabilities that the fixed-signature model does not have. The first is label switching [10]. The signature index has no fixed meaning, so any relabeling of the K signatures, together with the matching relabeling of the activities, leaves the likelihood unchanged. The second is the trade off between signatures and activities. Only the product $e_j S$ enters the likelihood, so a range of pairs (S, e) can fit the data comparably well.

Label switching is handled outside the model. Every posterior draw is permuted to a reference labeling, the mean of one chain, by maximum cosine similarity, a linear assignment problem [7]

that matches each inferred signature with a reference signature to maximise the total cosine. The alignment permutes only the signatures and the aligned activities, so convergence is judged on those variables. The raw walk variables η and z are not aligned, since their last axis has length $K - 1$ and a K -permutation does not apply to them cleanly, so their $\hat{r}$ is inflated by label switching and is excluded from the assessment.

4.3 The chains settle into several clusters

The dataset is a single moderate overlap forest, twenty trees over ten signatures generated at correlation $\rho = 0.3$, sampled with eight chains. Aligned to a common labeling, the chains do not agree, settling into several clusters that fit the data about equally well. By their mean activity fingerprint they fall into a coarse split of five chains against three, cluster A and cluster B, though the between chain view below is finer than two.

The disagreement is confined mainly to one signature. Figure 2 scores every chain against the true signatures by Hellinger distance, a distance between distributions sensitive to the small channels where similar signatures differ. Nine of the ten are returned with every chain within a few hundredths of the others. The exception is signature 2, the least used. A signature’s exposure is its total activity across the forest, counted in node-equivalents, one node-equivalent being a node that uses that signature and nothing else. By that measure signature 2 sits at about 2, against forty for the abundant signatures. There the clusters sit 0.10 apart, several times more than any other. The chains agree on what the data constrain well and differ only in the directions the data constrain weakly.

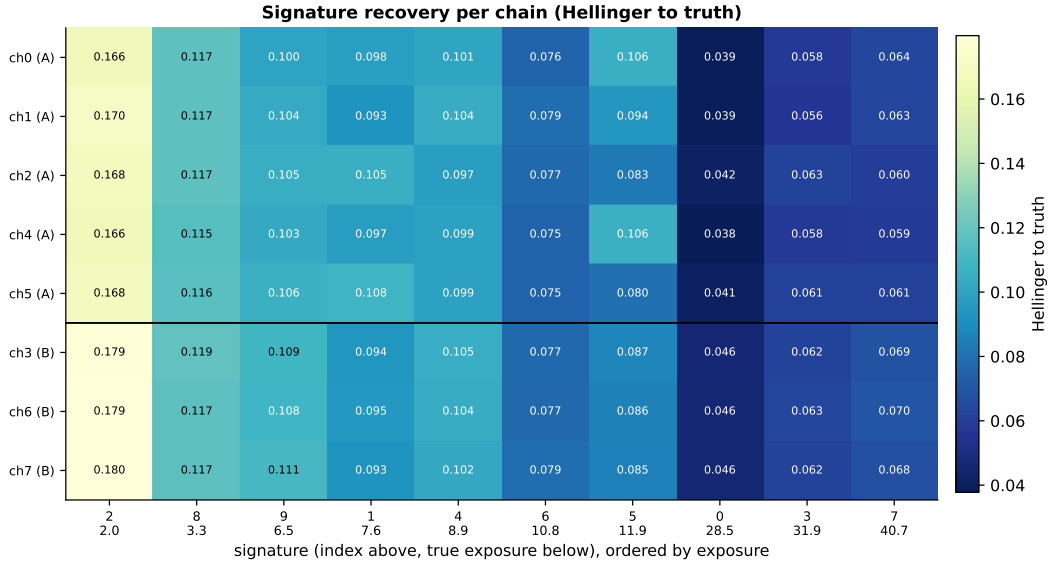


Figure 2: Per chain signature recovery, each cell the Hellinger distance from a chain’s reconstruction to the truth, columns ordered by exposure and rows grouped by cluster. The chains agree down every column except signature 2, the least used, where the rows separate by cluster.

Comparing the chains to each other rather than to the truth shows the clusters directly. Figure 3 gives the between chain distance for the split signature and an abundant control, in signature shape and in activity. On the rare signature the chains break into separated blocks finer than the coarse two, three forming one tight cluster within 0.01 of one another and the other five splitting further. The control, by contrast, is a single uniform block in both panels.

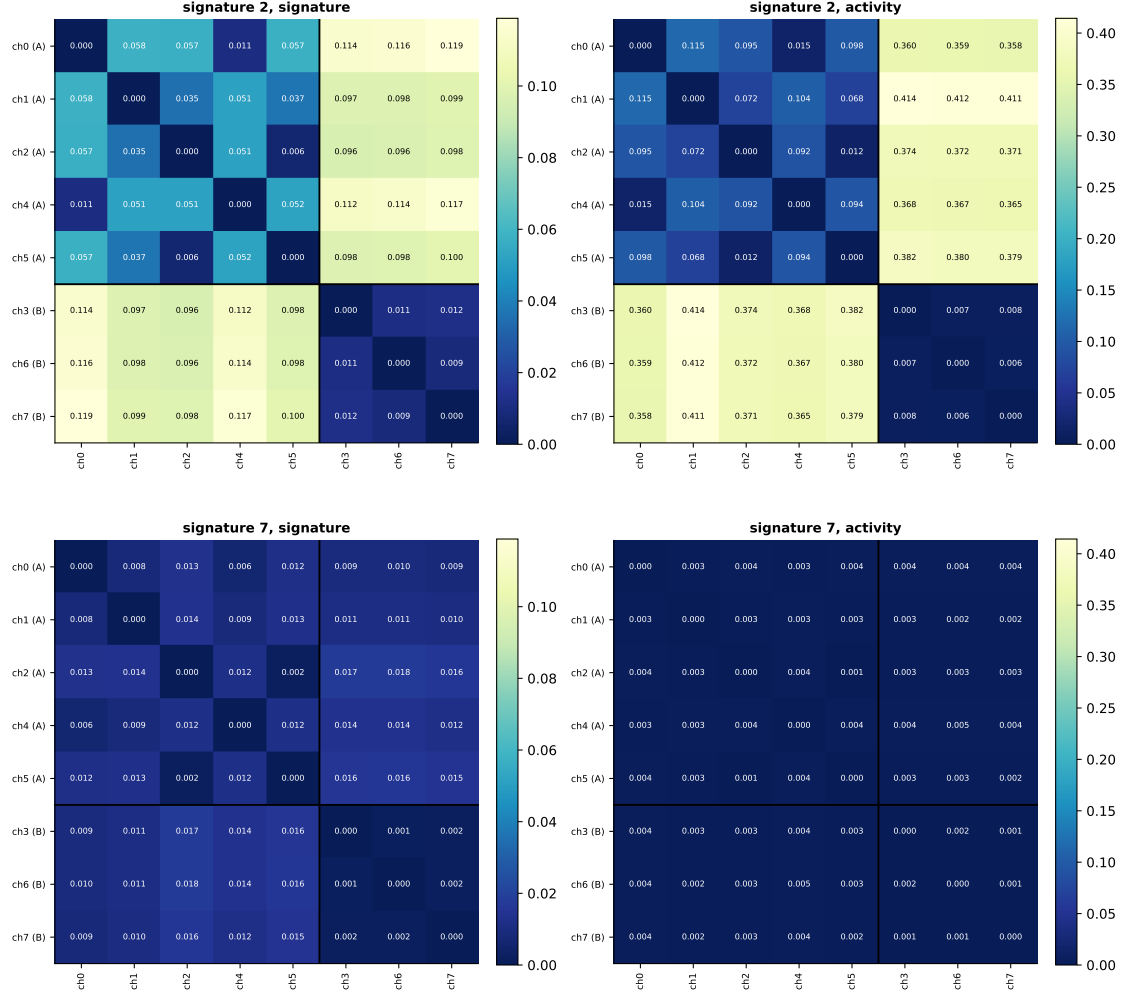


Figure 3: Between chain distance on the split signature 2 (top) and an abundant control (bottom), in signature shape (left, Hellinger) and activity (right, Bray-Curtis), chains ordered by cluster, colour scale shared down each column. On the rare signature the chains break into blocks finer than the coarse two. The control is a single uniform block.

Figure 4 reconstructs signature 2 under each cluster against the truth, cluster A at Hellinger 0.17 and cluster B at 0.18, and 0.10 apart from each other. That is eight times the 0.012 between the clusters on the abundant control, and more than half as large as either cluster’s own distance to the truth, so the two recover different shapes for the rare signature.

The activity differs as well, and the two are tied together. Because only the product $e_j S$ enters the likelihood, the rare signature’s shape and its activity are pinned only jointly, so a cluster can offset a change in one against the other. Figure 5 compares the per node activity each cluster assigns to signature 2. Cluster B places more at nearly every one of the 152 nodes, about twice as much in total, 4.85 against 2.22 node-equivalents, against a true exposure of about 2. The activity gap is the wider of the two, but the split is in the pair, not in either alone.

The non-convergence therefore sits in a narrow part of the posterior, the rarest signature’s reconstruction, its shape and activity together, rather than across the whole of it. In this run that is close to a single direction, since one signature is rarer than the rest.

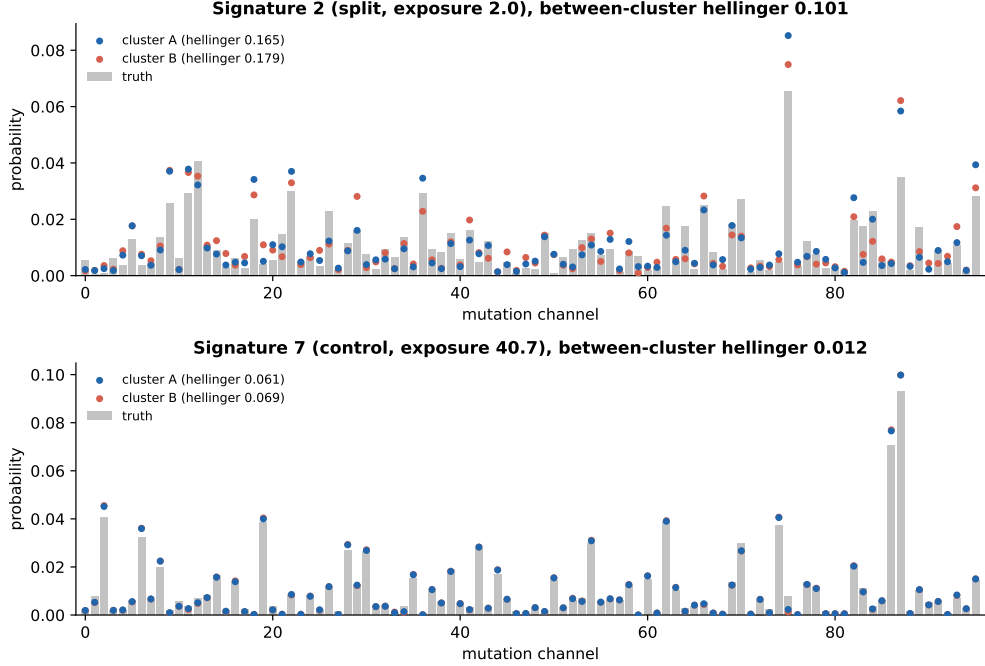


Figure 4: Signature 2 (top) and an abundant control (bottom) under each cluster (markers) against the truth (bars). The clusters differ on the rare signature, cluster A at Hellinger 0.17 and cluster B at 0.18, but coincide on the control.

4.4 Mixing geometry

Before measuring the distance between the clusters we align one cluster to the other, which removes any permutation of the signature labels and leaves only that genuine difference. We then join the two representatives by a straight line through all the parameters, the activities and the signatures together, and evaluate the log-likelihood and log-posterior at each point along it. Figure 6 shows the result.

Along the line neither the likelihood nor the posterior dips between the clusters. Both rise into a hill in the middle, the likelihood by 379 nats, so the path from one cluster to the other runs over high ground rather than through a valley. The posterior does tilt toward cluster A, but which cluster it prefers is beside the point. What matters for mixing is that nothing along the line pens the chains into one cluster. The clusters are joined by a ridge, and a sampler free to move along it would visit both within a single chain.

The chains nonetheless stay apart, and the reason is geometric. At each point on the straight line taken in the sampler’s own coordinates we measure the curvature of the log posterior, its second directional derivative, along the line and across it. Along the line the curvature is soft, with median magnitude 1.1. Across it the walls are stiff. The stiffest curvature in any direction is the largest magnitude eigenvalue of the Hessian of the log posterior. We never build that Hessian, which at this model’s size would be far too large. Power iteration needs only the Hessian applied to a vector, and each such Hessian-vector product is computed directly by the method of Pearlmutter [9] without forming the matrix. Each product stretches the vector further along the steepest direction, until it points only that way and no other, and the curvature there is about 1.5×10^5 .

Their ratio, the condition number, is then about 1.3×10^5 , a steep walled canyon whose soft floor runs the length of the line joining the clusters and lies along the activity of the rarest signature,

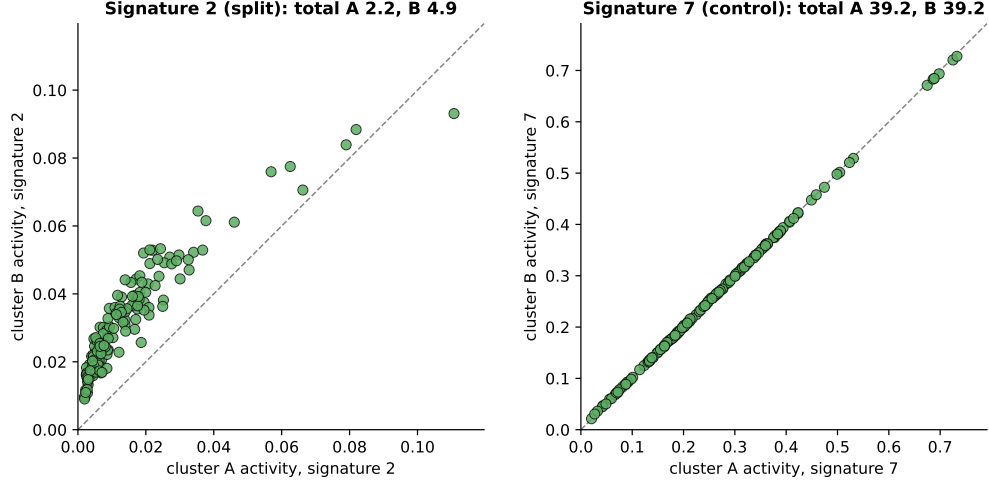


Figure 5: Per node activity assigned to each cluster, signature 2 (left) and an abundant control (right). On signature 2 cluster B sits above the diagonal at almost every node, 4.85 against 2.22 node-equivalents in total against a true exposure of about 2. On the control the clusters lie on the diagonal.

the one the data constrain least.

This anisotropy, the gap between the soft and stiff curvatures, is what stalls the sampler. The stiffest curvature sets the largest step the leapfrog integrator can take and stay stable, two over its square root, which here is about 0.005. The sampler adapted a step of 0.0023, about half of that. The line between the clusters is long in these coordinates, an arclength of 62, the length of the path in the sampler's units, and a step that small needs about 2.7×10^4 leapfrog steps to travel it from one cluster to the other. The longest trajectory the sampler builds at its tree depth of 10 is $2^{10} = 1024$ steps, so a single trajectory covers only a small fraction of the distance. The sampler is not stopped by a wall in the value but held by a floor too long to walk in the time the integrator is allowed. This is why the signature of the trap is a saturated tree depth and a small step size with no divergences. The integrator stays stable and accurate, and is simply too slow.

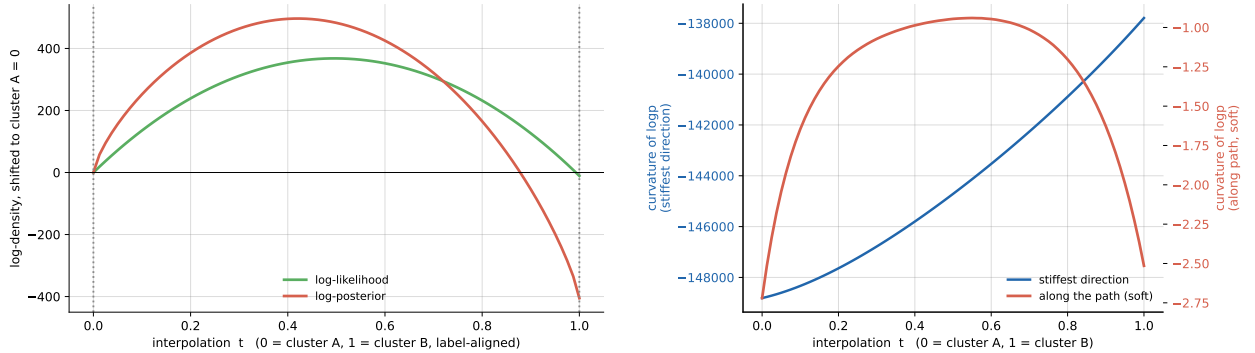


Figure 6: Mixing geometry of the cluster split. Left, model log likelihood and log posterior along the straight line between the two clusters. The likelihood is tied at the ends and humps between them, while the posterior tilts to cluster A. Right, the curvature along the path and the stiffest curvature across it, soft along and stiff across.

4.5 The reparameterisation

The diagnosis points at two faults, a soft direction the clusters slide along and the anisotropic geometry around it. The soft direction is the rarest signature’s reconstruction. Signature 2 carries so little exposure that the data barely fix the product $e_j S$ there, so its shape and its activity can trade off against each other with almost no change to the fit. In the plain walk the signature’s overall activity level is never represented on its own. It is built up node by node and tied down only at the root, so it drifts, and the two clusters are two settlements of that drift, cluster A placing 2.22 node-equivalents on signature 2 and cluster B 4.85 against a true exposure of about 2 in our example (Figure 5). We add an explicit forest pooled usage level per signature, drawn from a sum-zero normal and shared across the whole forest, with each tree root deviating from it in a non-centred form. The level is then a single parameter that every node informs, so the rarest signature’s usage is pinned by that pooled estimate rather than left to drift, and with its activity fixed the coupled shape is fixed with it.

The geometry is the second fault. The walk maps its logits to the simplex with a softmax whose last coordinate is pinned to zero, a reference signature the map needs to be identifiable. Measuring every other signature as a ratio against that one reference is neither symmetric nor isometric. A step of a given size moves different signatures by different amounts, and the coordinates are coupled through the reference. The soft level direction is then aligned with no coordinate and not orthogonal to the stiff shape directions. That is the anisotropy the curvature picked up, and a diagonal step size cannot rescale it. We replace the anchored softmax with a symmetric sum-zero link. The logits are constrained to sum to zero and mapped to the simplex with no pinned coordinate, so every signature is treated alike and a step is the same size in every direction.

Written together, the two changes give the activity prior

$$\mu \sim \mathcal{N}_0(\mathbf{0}, \sigma_\mu^2 I), \quad z_{0,t}, z_j \sim \mathcal{N}_0(\mathbf{0}, I) \quad (7)$$

$$\eta_{\text{root}(t)} = \mu + \sigma_0 z_{0,t}, \quad \eta_j = \eta_{\text{parent}(j)} + \sigma z_j \quad (8)$$

$$e_j = \text{softmax}(\eta_j) \quad (9)$$

Here $\mathcal{N}_0$ is a Gaussian on the sum-zero subspace, so the logits sum to zero, realised through an isometric log-ratio basis so the increments stay standardised, with σ under a $\text{LogNormal}(0, 1)$ prior and the scales $\sigma_0 = 1$ and $\sigma_\mu = 2$ fixed. The softmax now takes the full sum-zero logit with no pinned coordinate. The forest-pooled level μ enters once for the whole forest, and each tree root t deviates from it through $\sigma_0 z_{0,t}$. The signature prior and the count likelihood are unchanged from the de novo model.

After applying both fixes the chains no longer fall into clusters. Next we will look at the recovery and compare it with NMF.

4.6 What makes recovery hard

We will focus on two properties of a forest and signatures that can influence how hard it is to recover activities and signatures. The first is signature overlap. Two similar signatures trade off against each other with almost no change to the predicted spectrum, so the activity difference between them is nearly invisible to the data. Here, the signatures are generated at a tunable correlation ρ , which sets this overlap directly. The second is usage. A rare signature, one with low exposure, leaves few mutations from which to estimate its activity. Exposure increases as the forest grows. The two sweeps that follow vary signature correlation and the number of trees.

4.7 Recovery against NMF

With the clusters removed, we ask how well the de novo model recovers the truth and how it compares to NMF, the standard signature tool, which treats the nodes as exchangeable and discards the tree, running on the observed node count matrix with the scikit-learn implementation and its generalised Kullback-Leibler loss. Both run with the true number of signatures, Tree-HDP with eight chains and NMF as the best of ten restarts by reconstruction error. Both are scored against the generating signatures and activities, signatures by Hellinger distance and activities by the L_1 distance, lower being better, averaged over fifteen independent forests at each setting.

We vary the two difficulty axes in turn, each holding the other fixed. The size sweep grows the forest from five to sixty trees at a fixed signature correlation $\rho = 0.3$, with one signature set shared across every size so that the number of trees is the only variable. The overlap sweep fixes the forest at twenty and forty trees and moves ρ from 0.1 to 0.9, drawing a new set of signatures at each ρ (Figure 7).

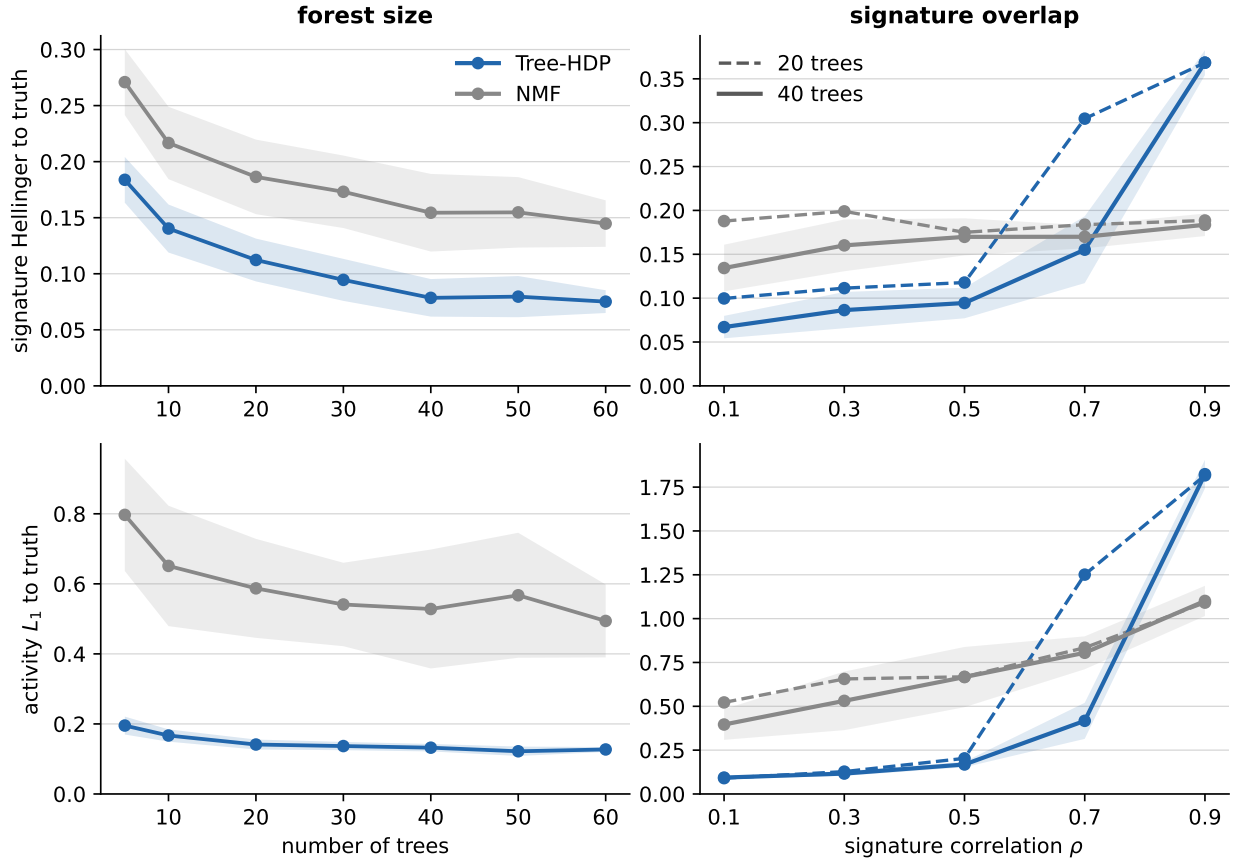


Figure 7: Recovery of Tree-HDP against the NMF baseline, signatures by Hellinger (top) and activities by the L_1 distance (bottom), lower being better. The left column is the size sweep, Tree-HDP against NMF. The right column is the overlap sweep, overlaid at twenty trees (dashed) and forty trees (solid), with $\pm$ standard deviation bands over fifteen forests shown for the size sweep and the forty-tree curves. Doubling the forest lowers Tree-HDP’s error at $\rho = 0.7$ and moves the crossover with NMF to higher overlap, but leaves $\rho = 0.9$ unchanged.

Across forest size Tree-HDP recovers both signatures and activities better than NMF at every

size, and the activity gap is the wider one. It holds across the range, an L_1 around 0.14 for Tree-HDP against 0.5 to 0.8 for NMF, since sharing activity between related nodes is what the tree prior adds and NMF discards. Both methods improve as the forest grows and the gap narrows only slowly, so the advantage is a standing one rather than a small sample effect. Tree-HDP is also the steadier of the two, its spread across forests several times tighter than NMF's.

Across signature overlap the picture is not a clean win, and it depends on how much data the forest carries. At twenty trees, while the signatures are distinguishable, $\rho \leq 0.5$, Tree-HDP is far ahead, recovering activities at an L_1 around 0.09 to 0.20 against NMF's 0.52 to 0.67. Past $\rho = 0.5$ it degrades and crosses below NMF, and by $\rho = 0.9$ it is the worse of the two, signature Hellinger 0.37 against 0.19 and activity L_1 1.82 against 1.10. This is not a sampling failure. Convergence stays clean across the sweep, with $\hat{r}$ below 1.01 and healthy ESS even at $\rho = 0.9$, so the chains agree and still miss the truth.

Doubling the forest to forty trees separates two regimes. At $\rho = 0.7$ the extra data rescues Tree-HDP, its signature error falling from 0.31 to 0.16 and its activity L_1 from 1.25 to 0.42, so it moves from behind NMF to ahead of it and the crossover shifts to higher overlap. At $\rho = 0.9$ the extra data changes nothing, signature Hellinger and activity L_1 hold at 0.37 and 1.82. When the signatures overlap this heavily the model converges cleanly to a smooth, well-fitting solution, the data cannot pull toward the truth. NMF gains modestly from the extra data as well, but the tree prior helps most where the signatures can still be told apart.

5 Discussion

Every result here was measured under idealised conditions, and each idealisation is something a real cohort would break. The number of signatures K was held at its true value, whereas on real data it is unknown, so unknown- K inference, where the label and identifiability problems seen here compound with not knowing how many signatures to seek, is the natural next stage. The mutation counts are split across channels by a multinomial in both the generator and the inference, so that split carries only sampling noise, and a Dirichlet-multinomial channel model would add overdispersion. Every conclusion also rests on distance to a known truth, which is not known for real data. For testing on real data we need other evaluation methods such as stability across seeds, or agreement with reference catalogues. The phylogeny is taken as fixed and correct, while in practice it is reconstructed upstream and itself uncertain. Topology errors would feed straight into the activity inference. Finally the activity prior carries a single global step size, one rate of drift for every edge, where a richer prior would tie the change across an edge to its length and let the similarity between nodes fall off with their distance on the tree.

References

- [1] Oriol Abril-Pla, Virgile Andreani, Colin Carroll, Larry Dong, Christopher J. Fannesbeck, Maxim Kochurov, Ravin Kumar, Junpeng Lao, Christian C. Luhmann, Osvaldo A. Martin, Michael Osthege, Ricardo Vieira, Thomas Wiecki, and Robert Zinkov. PyMC: A modern, and comprehensive probabilistic programming framework in Python. *PeerJ Computer Science*, 9: e1516, 2023. doi: 10.7717/peerj-cs.1516. URL <https://doi.org/10.7717/peerj-cs.1516>.
- [2] Ludmil B. Alexandrov, Serena Nik-Zainal, David C. Wedge, Peter J. Campbell, and Michael R. Stratton. Deciphering signatures of mutational processes operative in human cancer. *Cell*

- Reports*, 3(1):246–259, 2013. doi: 10.1016/j.celrep.2012.12.008. URL <https://doi.org/10.1016/j.celrep.2012.12.008>.
- [3] Ludmil B. Alexandrov, Serena Nik-Zainal, David C. Wedge, et al. Signatures of mutational processes in human cancer. *Nature*, 500(7463):415–421, 2013. doi: 10.1038/nature12477. URL <https://doi.org/10.1038/nature12477>.
 - [4] Ludmil B. Alexandrov, Jaegil Kim, Nicholas J. Haradhvala, et al. The repertoire of mutational signatures in human cancer. *Nature*, 578(7793):94–101, 2020. doi: 10.1038/s41586-020-1943-3. URL <https://doi.org/10.1038/s41586-020-1943-3>.
 - [5] Matthew D. Hoffman and Andrew Gelman. The no-u-turn sampler: Adaptively setting path lengths in Hamiltonian Monte Carlo. *Journal of Machine Learning Research*, 15(47):1593–1623, 2014. URL <https://jmlr.org/papers/v15/hoffman14a.html>.
 - [6] Katharina Jahn, Jack Kuipers, and Niko Beerenwinkel. Tree inference for single-cell data. *Genome Biology*, 17:86, 2016. doi: 10.1186/s13059-016-0936-x. URL <https://doi.org/10.1186/s13059-016-0936-x>.
 - [7] Harold W. Kuhn. The Hungarian method for the assignment problem. *Naval Research Logistics Quarterly*, 2(1–2):83–97, 1955. doi: 10.1002/nav.3800020109. URL <https://doi.org/10.1002/nav.3800020109>.
 - [8] Daniel D. Lee and H. Sebastian Seung. Learning the parts of objects by non-negative matrix factorization. *Nature*, 401(6755):788–791, 1999. doi: 10.1038/44565. URL <https://doi.org/10.1038/44565>.
 - [9] Barak A. Pearlmutter. Fast exact multiplication by the Hessian. *Neural Computation*, 6(1):147–160, 1994. doi: 10.1162/neco.1994.6.1.147. URL <https://doi.org/10.1162/neco.1994.6.1.147>.
 - [10] Matthew Stephens. Dealing with label switching in mixture models. *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, 62(4):795–809, 2000. doi: 10.1111/1467-9868.00265. URL <https://doi.org/10.1111/1467-9868.00265>.
 - [11] Yee Whye Teh, Michael I. Jordan, Matthew J. Beal, and David M. Blei. Hierarchical Dirichlet processes. *Journal of the American Statistical Association*, 101(476):1566–1581, 2006. doi: 10.1198/016214506000000302. URL <https://doi.org/10.1198/016214506000000302>.
 - [12] Aki Vehtari, Andrew Gelman, Daniel Simpson, Bob Carpenter, and Paul-Christian Bürkner. Rank-normalization, folding, and localization: An improved R-hat for assessing convergence of MCMC. *Bayesian Analysis*, 16(2):667–718, 2021. doi: 10.1214/20-BA1221. URL <https://doi.org/10.1214/20-BA1221>.